Digital Vector · VCE Applied Computing | Unit 1 · Area of Study 1 · Data Analysis · KK02

Characteristics of data types

Every value you store has a type — and the type you choose decides what you're allowed to do with it. Pick the wrong one and your phone numbers lose their leading zeros, your dates become nonsense, and your totals quietly break.

What this key knowledge covers
This summary distils everything you need for **KK02 — Characteristics of data types** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

A data type is defined by what you can do with it

It's tempting to think a data type is just "what the value looks like". It isn't. The VCAA glossary defines a data type as the forms data can take, **characterised by the kind of operations that can be performed on it**. That single idea answers almost every exam question on this dot point.

You can concatenate (join) strings, search inside them, and pull out characters. You *cannot* meaningfully add or average them.

You can do arithmetic, find averages, sort by size and compare magnitudes. Choose this only when the maths actually means something.

Holds only TRUE or FALSE. Built for logic and filtering — "show me sightings where expert_confirmed = TRUE ".

The same value can be stored differently depending on whether you're in a **spreadsheet** or a **database**. The skill VCAA wants is choosing a type that suits both the *data* and the *tool* — and being able to justify it with the operations you need.

An analogy: kitchen containers

Think of data types as containers in a kitchen. A measuring jug is built for liquids you'll *pour and measure* (numeric). A labelled spice jar holds a *name* you'll read, never pour (text). A light switch is simply on or off (Boolean). You *could* pour milk into a spice jar — but you've lost the ability to measure it. Choosing a number type for a phone "number" is exactly that mistake: it fits, but you've thrown away what made the container useful.

The type sorter

Each chip below is a real value from the CritterCount dataset. Send it to a bin and watch what operations that type unlocks. The goal isn't "which bin will accept it" — it's "which bin lets you do the job you need".

Text — character and string

Text data is stored as characters. The study design names two forms:

A **single** symbol — one letter, digit, or punctuation mark. In CritterCount, a size_band field might store S, M or L — exactly one character.

A **sequence** of characters of any length — words, sentences, codes. species = "Eastern Grey Kangaroo" is a string. Most real-world text is a string.

Numeric — integer, floating point, date/time

Numeric types are for values you'll actually do maths with. The study design lists three sub-types.

Whole numbers, no decimal point. count_seen = 7. You can't see half a kangaroo, so a count is an integer.

Numbers with a decimal fraction. air_temp_c = 14.6, or a GPS latitude like -37.8136. Use it whenever precision below 1 matters.

A special numeric type. sighted_at is stored as a number under the hood, so you can sort chronologically and subtract two dates to get a duration.

If you stored 14/03/2026 as plain text, sorting would put 14/03 before 2/04 alphabetically — wrong order. A real date/time type stores it as a number, so "earliest first" actually works and you can calculate "days since first sighting".

Boolean — true or false, nothing else

A Boolean holds one of exactly two values: TRUE or FALSE (sometimes shown as Yes/No, 1/0, or a tick-box). In CritterCount, expert_confirmed is Boolean — a sighting either has been verified or it hasn't.

Why not just store the word "yes" as text? Because Boolean unlocks **logical operations**. You can filter WHERE expert_confirmed = TRUE, combine conditions with AND/OR, and count flagged records instantly. Text "yes"/"Yes"/"y"/"YES" would be inconsistent and far harder to filter reliably.

If a question can only ever be answered *yes* or *no* — "Is it nocturnal?", "Was a photo attached?", "Confirmed?" — it's a Boolean. The moment a third option appears ("maybe", "pending"), it's no longer Boolean.

The same value, matched to the chosen software tool

This dot point specifically says data types are matched to the *chosen software tools*. A spreadsheet and a relational database name their types differently, even for the same value. Knowing the mapping is worth marks.

If a question names a tool ("…in the spreadsheet" or "…in the database"), use that tool's type names and justify the choice by the operations the data needs. Don't just say "number" — say "*Number formatted with no decimal places, because counts are whole values that will be* *summed*".

Number or text? Make the call

For each CritterCount field, decide whether it should be stored as a number or as text. The test is always the same: **will you do arithmetic with it?** If not, it's text — even when it's made of digits.

Six exam traps

Phone, postcode, observer ID, ABN, credit-card — digits, but never added. Store as **text/string** to keep leading zeros and full length.

The 2023 report noted students lost higher marks because they stated the rule but gave **no concrete example**. Always anchor with one: "e.g. a phone number 0431… loses its leading zero".

A **character** is exactly one symbol; a **string** is many. A size band M is a character; the species name is a string.

Text dates sort alphabetically and can't be subtracted. Use a real **date/time** type so chronological sorting and date arithmetic work.

"A number is something with digits" earns little. Define by **the operations allowed** — that's the glossary definition and the mark-winning angle.

If the question says spreadsheet or database, use **that tool's type names** (Short Text, Number, Yes/No), not generic words alone.

The P·E·L writing trainer

For any "justify the data type" question, structure the answer as **Point → Evidence → Link**. It forces in the example examiners reward.

"Store observer_id as **text (Short Text in the database)**."

"IDs are never used in calculations, and a numeric type would drop the leading zero — e.g. 00742 would become 742."

"So every ID stays complete and unique, which the matching of observers to sightings depends on."

"Store observer_id as text. IDs aren't used in calculations and a numeric type would drop the leading zero (00742 → 742), so storing it as text keeps every ID complete and unique for matching records." — that's a full-mark *justify* response in three sentences.

Practice questions

SAC/exam-style questions in VCAA command-word style. Try each in your head or on paper, then reveal a weak answer, a strong answer and the marking guide. Award yourself marks — the dial bottom-right tracks your total. (Units 1 & 2 are school-assessed, so these mirror the style your teacher will use.)

Total available: **15 marks**. Be honest with the marking guides — examiners reward the named example and the lost operation, not just the right type word.

Recap

- A **data type** is defined by the **operations** you can perform on it — not by how the value looks.
- Text : **character** (one symbol) and **string** (many). For values you read, join or search — never add.
- Numeric : **integer** (whole), **floating point** (decimal), **date/time** (numeric under the hood). For values you calculate, compare or sort.
- Boolean : TRUE/FALSE only, for logical filtering.
- Match the type to the **software tool** (spreadsheet vs database) and use that tool's type names.
- The big trap: **digit-strings you never calculate with** (phone, postcode, ID) are **text** — numeric drops leading zeros.
- In a *justify*, always give a **concrete example**. It's the difference between mid and full marks.

Key terms

Term	Meaning
State / Identify	Name the data type. One or two words. "Integer." No justification needed — but don't write a paragraph for a 1-mark name.
Describe	Say what the type is *and* a defining feature or operation. "Boolean — stores only TRUE or FALSE, used for logical filtering."
Explain / Justify	Give the choice, the reason (the operations needed), *and* a concrete example. This is where the leading-zero example earns the top mark.
Compare	Put two types side by side on the same trait — e.g. integer vs floating point on whether decimals are stored.

Common traps & misconceptions

Exam trap
⚠ The leading-zero trap (this is the one) CritterCount gives every observer an ID like 00742. Phone numbers, postcodes and member IDs are the same. They *look* like numbers but you never add them — and storing them as numeric silently deletes the leading zero, turning 00742 into 742. **Rule:** if the "number" isn't used in calculations — phone, postcode, ID, ABN — store it as **text/string**. The interactive below makes the zero physically fall off.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define · 2 marks

The term *data type* and give one example of a data type.

✗ A weak answer

· 0–1 "A data type is the type of data. For example a number." Circular ("type of data"), and misses the defining idea. The example is acceptable but the definition earns nothing.

✓ A strong answer

· 2/2 "A data type is the form a value can take, defined by the operations that can be performed on it. For example, an *integer* stores whole numbers that can be added or compared."

How the marks are awarded

- 1 mark — definition referring to the form of data *and* the operations performed on it.
- 1 mark — one valid example (text/string, character, integer, floating point, date/time, Boolean).

Q2. Justify · 3 marks

CritterCount stores each observer's mobile number, e.g. 0431 251 902. the most appropriate data type for this field.

✗ A weak answer

· 1 "It should be a number because a phone number is a number." States a choice but the reasoning is the trap. No mention of operations, leading zeros, or an example of what goes wrong.

✓ A strong answer

· 3/3 "Store it as **text (string)**. A phone number is never used in calculations, so the arithmetic operations of a numeric type aren't needed. A numeric type also drops leading zeros — 0431… would become 431… — so storing it as text keeps the number complete and dialable."

How the marks are awarded

- 1 mark — correct type: text / string.
- 1 mark — reason tied to operations (no arithmetic needed) or formatting.
- 1 mark — concrete consequence/example (leading zero dropped, e.g. 0431431).

Q3. Distinguish · 2 marks

Between the *character* and *string* data types, using a CritterCount field for each.

✗ A weak answer

· 1 "A character and a string are both text. A string is longer." "Longer" gestures at the idea but isn't precise, and there are no fields used as required.

✓ A strong answer

· 2/2 "A **character** stores exactly one symbol — e.g. the size_band field holding M. A **string** stores a sequence of characters of any length — e.g. species holding "Eastern Grey Kangaroo"."

How the marks are awarded

- 1 mark — character = single symbol; string = sequence of characters.
- 1 mark — a correct, distinct example field for each.

Q4. State · 4 marks

The CritterCount team will record count_seen, air_temp_c, sighted_at and expert_confirmed in their **spreadsheet**. the most appropriate data type for each and briefly say why.

✗ A weak answer

· 2 "count = number, temp = number, date = date, confirmed = text." Two slips: it doesn't distinguish integer vs floating point, and expert_confirmed should be Boolean, not text. No reasons given.

✓ A strong answer

· 4/4 count_seen — **integer** (Number, 0 decimals): whole animals, summed. air_temp_c — **floating point** (Number with decimals): needs fractional precision. sighted_at — **date/time**: enables chronological sorting and date arithmetic. expert_confirmed — **Boolean** (TRUE/FALSE): two-state flag for logical filtering.

How the marks are awarded

- 1 mark each (4 total) for the correct type with a valid reason. Integer/floating-point distinction required; Boolean (not text) required for the flag.

Revision checklist

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- Text : character (one symbol) and string (many). For values you read, join or search — never add.
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- Boolean : TRUE/FALSE only, for logical filtering.
- Match the type to the software tool (spreadsheet vs database) and use that tool's type names.

- The big trap: digit-strings you never calculate with (phone, postcode, ID) are text — numeric drops leading zeros.
- In a justify, always give a concrete example. It's the difference between mid and full marks.